
Postgres Performance Tuning Guide

Where to look first

pg_stat_statements ranked by total time, then the Grafana dashboards for cache hit ratio, checkpoint frequency, and replication lag. Most regressions trace to a new query pattern, a missing index, or autovacuum falling behind (see postgres vacuuming).

Connection layer

PgBouncer in transaction mode; keep active connections under $2 \times$ cores. Pool exhaustion shows as latency cliffs, not errors — the connection pooling doc has the sizing worksheet. Long transactions block vacuum: alert at five minutes.

Query patterns

- Prefer covering indexes for the invoice processing hot paths; verify with EXPLAIN (ANALYZE, BUFFERS).
- Batch writes through the Kafka consumer to avoid row-at-a-time churn.
- Partition the events table monthly; drop partitions per the data retention policy instead of DELETE.

Replication

Keep lag under two seconds (the disaster recovery RPO depends on it). Rebuild bloated replicas from database backups rather than letting them limp — the backup verification drill doubles as the rehearsal.